Class 10: The portfolio approach and the Price of securities

- Review of portfolio approach
- Problems with the portfolio approach
- CAPM
- Efficient market hypothesis
- Rise of index funds and passive investor;

Key concepts

- Sharpe optimal portfolio
- Capital asset pricing model
- Representative agent equilibrium
- Systematic risk
- Diversifiable risk

The efficient frontier

- As long
  - Investors only care about some statistic related to the mean and variance
  - Not say some percentile value of expected returns or about higher moments of the distribution
- Then set of portfolios that are on the efficient frontier is invariant to assumptions
    - Min $Var(x)$ conditional on $E(x) \geq r$
    - Max $E(x)$ conditional on $Var(x) < \sigma^2$
    - Both produce the same set of efficient portfolios
    - If you have a riskless asset, then unique Sharpe efficient portfolio
- Critical empirical issue
  - Forming expectations of return and variance going forward.

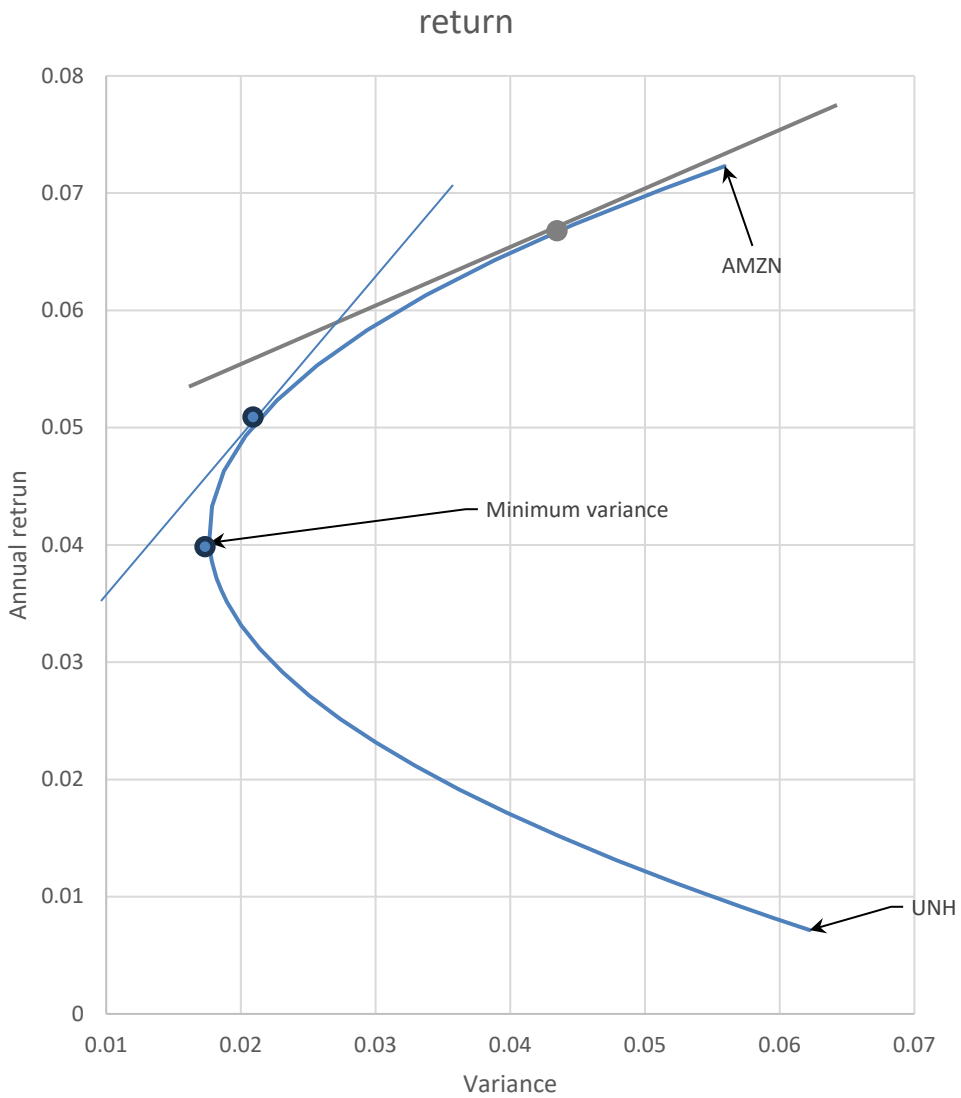
Optimal portfolio

- Take (your forecast) of the expected price and variance as given find optimal portfolio
- Prices come from demand and supply
  - How are they set?
- Variance comes from
  - a. the fundamental volatility of the NPV of asset
  - b. the volatility of demand for the asset
- For now we assume supply is fixed so we need to know demand.

Problem(s)

- The variance covariance matrix may be related to fundamentals
- Prices, however, (and thus returns) depend on demand
- Take all the optimal portfolios
  - Sum the asset quantity (e.x # of shares, or value)
  - That tells you aggregate demand.
  - Aggregate supply is the number of shares that company has issued.
- Equilibrium => for all the assets demand=supply?
- Or
  - Are some assets in excess supply
  - Are some assets in excess demand
- If so prices must adjust!

Example back to AMZN-UNH



	Return statistics	
	AMZN	UNH
Mean	0.076	0.007
Var	0.066	0.062
Covariance	-0.029	

	r	AMZN	UNH
Investor 1	5%	62.0%	38.0%
Investor 2	6.5%	83.8%	16.2%

Example (2)

- Demand
- # of investor of type 1 + # of investors of type 2

	r	AMZN	UNH
Investor 1	5%	62.0%	38.0%
Investor 2	6.5%	83.8%	16.2%
Share of investor 1			
25%		78.35%	21.65%
50%		72.90%	27.10%
75%		67.45%	32.55%

- Supply
- Market Cap AMZN 2589
- Market Cap UNH 255
- Market share UNH 9%
- So market equibrium fails
- Always excess demand for UNH

Lesson from the example

- If investors take prices as given and optimize their portfolios, there is not particular reason for the market to reach equilibrium
- What is the price formation rule?
- Solve this in two steps
  - Today mostly theory
  - Friday mostly algebra.

Solution-**CAPITAL ASSET PRICING MODEL**

- Investment market is defined by
  - One riskless security
  - A bunch of risky securities.
- Question
  - what are the prices of the risky securities?
  - what do portfolios look like?
- Hard to answer if there is a lot of heterogeneity (people all have different preferences)

Assumptions

- Assume large number of identical investors
- Everyone has mean-variance preference with the same parameters (a,b).
  - U(x) =R(x)+bVar(x)
- Many people gives us competitive prices
  - This is also an assumption of no private information
- Set up gives the two-fund separation theorem
  - Optimal portfolios are combinations of Sharpe optimal portfolio and riskless assets

Consequence of Assumptions

- If everyone is identical
  - They face the same prices
  - and have the same preferences
  - So they make the same choices
- In any equilibrium, everyone will have the same portfolio.
- And everyone must own the whole market.
  - Because if one investor does not want to hold a security then no one wants to.
  - This also implies no one goes short.
    - (because there is no one to hold the long position)

If Equilibrium then

- (1) All securities are allocated
  - Aggregate demand=aggregate supply
- (2) The weight of each security in the portfolio (w_i) has to be the ratio of the total value of that security divided by the total value of all securities. ($w_i = v_i/\Sigma V_j$)
- The weight is the proportional market cap.

=>The portfolio is a value-weighted index fund

Implication 2

- The market portfolio has to be the Sharpe optimal portfolio.
  - If the market is not the Sharpe optimal, then there has to be some assets that are in excess supply or demand
  - Best trade-off between risk and return conditional on the preferences of all our identical individuals.
  - At the market portfolio the tangent of the frontier is the marginal cost of risk.
  - If not, some assets are badly priced (some too expensive some too cheap)

Implication 3

- (4) The line from the riskless asset to the market portfolio is tangent to the efficiency frontier at the market portfolio.
  - Any choice has two funds (Riskless assets and market portfolio)
  - If there is heterogeneity then all optimal portfolios are on that tangent.
  - So having solved the large number of identical investors' problem we have also solved it for many different types of investors

Quiz time

CAPM -price of the fund

- Aggregate property
  - The tangency condition implies that every asset should earn at least the risk-free rate, if it earns less just drop the price
  - The optimal portfolio says that the prices take into account the value of diversification so the only thing left is aggregate volatility in the market.
    - A-bvar(market)
  - Investor has to be paid the risk-free rate plus a price for the volatility of the aggregate portfolio.

Risk Definition

- **Systematic (un-diversifiable) risk**: that part of the variance of an asset that is perfectly correlated with aggregate variation (so you can't offset it)
- **Diversifiable risk**: That part of the variance of an asset that can be offset by the volatility of another asset.
- Both are quantifiable (not uncertainty)

CAPM asset prices

- How is an individual asset priced?
  - If assets are efficiently priced, returns are ex-ante equalized.
  - So now the issue is the price of risk of that asset (of its volatility).
  - Market cares about the systematic risk.
- An asset that is highly correlated to the market does not help with variance so the market has to pay the investor a premium to hold it (it adds risk)
  - Low price high return
- An asset that is poorly correlated to the market helps with variance so the investor pays a premium to hold it (it lowers risk)
  - Higher price lower returns
- Next time how to set the premia

Implications of CAPM for investors

- Gives simple rules for making decisions—use a broad market portfolio of risky assets and then adjust risk by varying the quantity of riskless or very low-risk assets. Because the market adjusts the weights you do not need to worry about choosing the risk portfolios.
- It has economic logic (prices are formed as part of an equilibrium that takes in a risk return logic)
- Rise of value weighted index funds and the passive investor
- Prices take care of everything. If there is new information, don't rebalance because it all in the prices anyway.

Side bar

- Why we price equities
  - If equity in a firm has a positive price in most states of the world,
    - Bonds have a low probability of default can just do NPV.
  - If equity has zero price in many states of the world need to treat bonds as if they were equity
    - They are assets with means and variance

CAPM and Leverage

- Two fund separation depends on a riskless asset
- But if leverage is high enough, default is possible (net value of fund negative)
- Two solutions
  - Lender has a liquidation option
    - Buy the portfolio at its net value at any time,
  - Investor faces an increasing cost of debt as default becomes more likely.
  - In practice this has to do with the investor's credit rating
    - If AAA then you can be very highly leverage
    - If not then have to post collateral outside the fund.

Class 11: CAPM and Market Efficiency

- The Math of CAPM
- Securities priced from a regression
- Quantitative evidence
  - Rise of index funds
- Critique of CAPM
  - Arbitrage Pricing Theory